

Question Paper

Exam Date & Time: 27-Dec-2025 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU

THIRD SEMESTER B.TECH. DEGREE MAKEUP EXAMINATIONS - DECEMBER 2025

Data Structures [ECM_2125]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Write a function Smallest to find the smallest element in an array using pointer. Create a dynamically allocated array and read the values from keyboard in main. Display the result in the main function. (4)
- A)

Course Outcome: "CO1" / Cognitive Level: "Application"

- B) Show the advantage of using pointers to swap two values when it is passed as a parameter to a function. (3)

Course Outcome: "CO1" / Cognitive Level: "Understand"

- C) Determine and explain the functionality of the following code fragment and write the output of any sample input (3)
- ```
void funtion1(int a[], int n, int k)
{
 k = k % n;
 int temp[n];
 for (int i = 0; i < n; i++)
 temp[i] = a[(i + k) % n];
 for (int i = 0; i < n; i++)
 a[i] = temp[i];
}
```

Course Outcome: "CO1" / Cognitive Level: "Application"

- 2) Given the following sequence of letters and asterisks: EAS\*Y\*QUE\*\*\*ST\*\*\*IO\*N\*\*\* (4)
- A) Consider the queue data structure, supporting two operations insert and remove. Suppose that for the above sequence, each letter (such as E) corresponds to an insert of that letter into the queue and each asterisk (\*) corresponds a remove operation on the queue. Show the sequence of values returned by the remove operations.

Course Outcome: "CO2" / Cognitive Level: "Analysis"

- B) Write a C program to accept a list of integer values from the user and insert them in a stack if they are prime numbers. Display the final stack and handle all stack conditions properly. (3)

Course Outcome: "CO2" / Cognitive Level: "Application"

- C) Convert the given infix expression to prefix expression. Show the content of Stack at every step (3)
- $A * (B + C / D) - E ^ F$

Course Outcome: "CO2" / Cognitive Level: "Application"

- 3) Write a C program to implement a **doubly linked list** that performs the following operations: (4)
- A) 1. Insert a node at the **beginning** of the list.  
2. Insert a node at the **end** of the list.
- Display the list in **forward** and **reverse** order.

Course Outcome: "CO3" / Cognitive Level: "Application"

- B) Write a C program to delete the duplicate element from the linked list. (3)

Course Outcome: "CO3" / Cognitive Level: "Analysis"

- C) Implement a stack using a linked list in C with some Example. Write functions to perform stack operations: push, pop, peek and isEmpty using linked list. Demonstrate these operations by pushing multiple elements onto the stack, popping some elements, and displaying the stack's state after each operation while handling stack underflow properly. (3)

Course Outcome: "CO3" / Cognitive Level: "Application"

- 4) Given the following Binary Search Tree with nodes inserted in order: 50, 30, 70, 20, 40, 60, 80, 10, 25, 35, 45 (4)
- A) Explain the cases of node deletion in a BST:
- Deletion of a leaf node
  - Deletion of a node with one child
  - Deletion of a node with two children

Deletion of root node

Course Outcome: "CO4" / Cognitive Level: "Application"

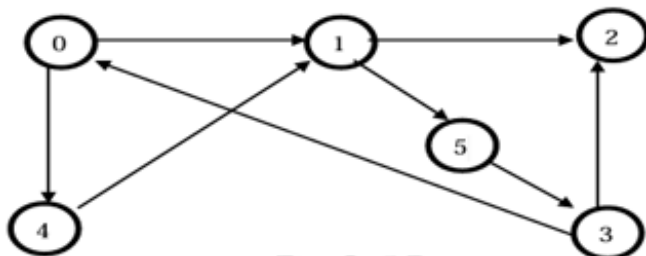
- B) Solve the following "A circular queue of size 6 initially contains elements [P, Q, R, S] with Front = 0 and Rear = 3. Perform the following operations: DEQUEUE(), ENQUEUE(T), ENQUEUE(U), DEQUEUE(), ENQUEUE(V), ENQUEUE(W) (3)
- (I) Show the status of the queue after each operation.  
(II) Indicate how Front and Rear pointers change after every operation.  
(III) Explain how the wrap-around mechanism avoids unused spaces.

Course Outcome: "CO2" / Cognitive Level: "Understand"

- C) Write and explain different graph representations. (3)
1. Adjacency Matrix
  2. Adjacency Lists
  3. Adjacency Multilists.

Course Outcome: "CO4" / Cognitive Level: "Understand"

- 5) Find the depth first and breadth first traversal for the graph shown in Figure. from all the vertices. (4)
- A) Write the function to find the depth first search taking as the parameter the value of the vertex from which DFS (depth first search) has to be carried out.



Course Outcome: "CO4" / Cognitive Level: "Application"

- B) Define the following with examples: (3)
- (I) Binary tree
  - (II) Complete binary tree
  - (III) Full binary tree
  - (IV) Binary search tree
  - (V) Depth of a tree

Course Outcome: "CO4" / Cognitive Level: "Understand"

- C) The following numbers are inserted into an empty binary search tree in the given order: 10, 1, 3, 5, 15, 12, 16. Create a BST. (3)
- What is the height of the binary search tree (the height is the maximum distance of a leaf node from the root)?

Course Outcome: "CO4" / Cognitive Level: "Understand"

-----End-----

| Event Name | Subject |
|------------|---------|
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058-MIT BLR-MAKEUP EXAM-2025-2026

ECM\_2125:Data Structures

| #            | Course Outcome | Description | Marks     |
|--------------|----------------|-------------|-----------|
| 1            | 1              | CO 1        | 10        |
| 2            | 2              | CO 2        | 13        |
| 3            | 3              | CO 3        | 10        |
| 4            | 4              | CO 4        | 17        |
| <b>Total</b> |                |             | <b>50</b> |

\* **NA** - CO Not Selected

| Event Name | Subject |
|------------|---------|
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058-MIT BLR-MAKEUP EXAM-2025-2026

ECM\_2125:Data Structures

| #            | Level       | No of Questions | Marks (%)        |
|--------------|-------------|-----------------|------------------|
| 1            | Remember    | 0               | 0(0%)            |
| 2            | Understand  | 5               | 15(30%)          |
| 3            | Application | 8               | 28(56%)          |
| 4            | Analysis    | 2               | 7(14%)           |
| 5            | Create      | 0               | 0(0%)            |
| 6            | Evaluate    | 0               | 0(0%)            |
| <b>Total</b> |             | <b>15</b>       | <b>50 (100%)</b> |